SSU Data Dash 2026: Team *Goats*

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Video (YouTube, unlisted is fine): `https://www.youtube.com/watch?v=FzzXqZb7SLY`
Code and submission files (GitHub): `https://github.com/UndefinedCpp/My-SSU-Data-Dash`

Summary

We build an automated valuation model for Toronto homes. It can evaluate the worth of a property before it is listed, and predict the sold price of a property on market. On the six-month walk-forward back test, our model achieved a median absolute percentage error of 6.94% off-market and 3.92% list-aware. Most of the performance gain came from careful feature engineering, with ensembling different algorithms providing a smaller additional reduction in error.

1 Data and features

The raw data was obtained from `https://huggingface.co/datasets/Baliyang/SSU_data_dash`. The training set has 150,000 sales, and there ares 3,974 off-market and 4,064 list-aware homes to test.

1.1 Data analyze

The training set comes with 22 columns. Only the external AVM columns are not used by our model. All other columns are used somewhere.

Column(s)	Usage
`type, style, beds, baths, parking, garage, area, lot front/depth, municipality, community, lat, lon, remarks, date_listed, days_on_market`	Used directly as features or to derive features
`list_price`	For list-aware track
`date_sold`	Training cutoff
`ml_num`	Row ID
`external_avm_estimate, external_avm_estimate_date`	Never used

The third-party AVM estimate is absent from both test files, and the estimate date comes after sales price in 71% of the listings. We dropped the two columns because a feature that is missing at prediction time and back-filled after the sale can hardly be informative or useful.

1.2 Features

Most features are derived, and we group them into 5 blocks. There are about 300 features in total.

1. **Base Features.** Base features are either directly pulled from the training set or simply derived from the original columns. For example, area midpoint; number of beds, baths, parking; latitude and longitude; number of months since 2021; six property-type flags; etc. Details can be found in code.
2. **Comparable Sales Features.** 14 features summarize the earlier sales near each home. To avoid leakage, only sales closed before the home was listed are used to build the features. Specifically, the features imply: what they sold for per square foot; how much they varied; how near and how recent they were; and whether local prices are rising.
3. **Building Features.** The intuition behind this feature block is that we have observed many sales with very similar, sometimes almost identical, geographic locations. Based on the assumption that those homes belong to the same building, we group the sales accordingly, identify individual buildings (clustering), and cross-reference them when encountered again.
4. **Space-time Spline Features.** The feature block is intended to capture space-time interactions. We assumed that (typical) price is a smooth, curved function of location and time. We applied cubic spline transformations to longitude, latitude and listing month, along with their tensor products. This generates 126 features in total.
5. **Text Features.** We extract signals from `remarks` column. We count 47 fixed terms into five cues ("renovated", "original", "potential", "luxury", "view"). Additionally, we employed a bag-of-words model that learns how much a wording moves the sale-to-list ratio, and a language model that generates the embeddings of the sentences.

2 Models

Off-market model is an ensemble of three:

- XGBoost regressor, trained on log price with squared error, sees all five feature blocks;
- CatBoost, trained on log price with square error, sees all features and raw remark as a bag of words;
- Semantic XGBoost regressor, trained on log price with absolute error, sees base features and the sentence embeddings.

List-aware model is much simpler. It is a single Histogram-base Gradient Boosting Regressor (HGBR) trained on log sold-to-list ratio with squared error. It consumes all features and an additional `historical_text_ratio` feature which is predicted by a ridge regressor on TF-IDF n-grams.

3 Validation

We score each track under two setups. The first is the six-month walk-forward back-test: for each month from December 2025 to May 2026, we retrain on every sale that closed before that month and predict the homes listed in it, pooling predictions and report metrics. The second holds out four months

at once: train through 31 January 2026, predict February to May. This mirrors the four-month horizon of the hidden test. Both assign homes to a month by listing date rather than sale date to avoid leakage.

4 Results

Track	MdAPE	MAPE	Median Error(%)	Mean Error(%)
Off-Market	6.9410, [6.758, 7.142]	9.3850, [9.182, 9.593]	+0.0471	+0.4001
List-Aware	3.9190, [3.821, 4.018]	5.0685 [4.974, 5.164]	−0.0621	−0.5705

Table 1: Walk-forward validation on the listings. MdAPE: median absolute percentage error. MAPE: mean absolute percentage error. Brackets: bootstrap 95% interval.

The six-month walk-forward is reported in Table 1. Off-market, the model reaches a MdAPE of 6.94%. Against the starter baseline of 20.8%, that removes 66.6% error from the baseline. List-aware, it reaches MdAPE 3.92% against a 7.3% baseline, which is 46% lower. Both tracks are close to unbiased in the median.

5 Where the model fails

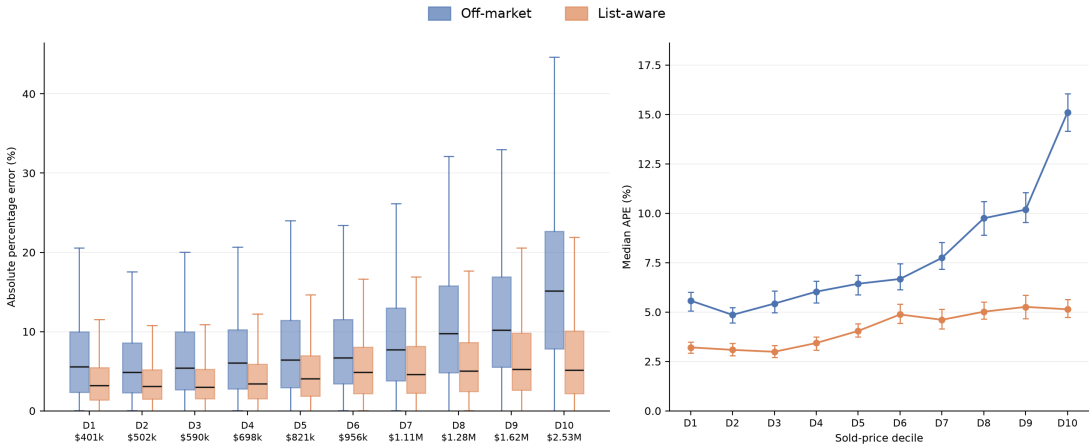


Figure 1: Absolute percentage error by sold-price decile.

In Figure 1 we plot absolute percentage error (APE) by sold-price decile, validated over February to May 2026 and trained on the sales before February. Left: APE distributions within each decile. Outliers are omitted for legibility, with whiskers extend to $1.5 \times IQR$. Right: median APE per decile with 95% bootstrap confidence intervals. Figure 2 shows signed percentage error by house type. Note that there are only 21 Co-Op Apartments, whose sample size is too small to be representative. Figure 3 shows signed percentage error by month.

6 Limitations

The models are weakest at the top of the market, where off-market APE reaches 15.1% in the highest price decile and under-predicts semi-detached and detached homes by roughly 3$\tilde{6}$%. These gaps are

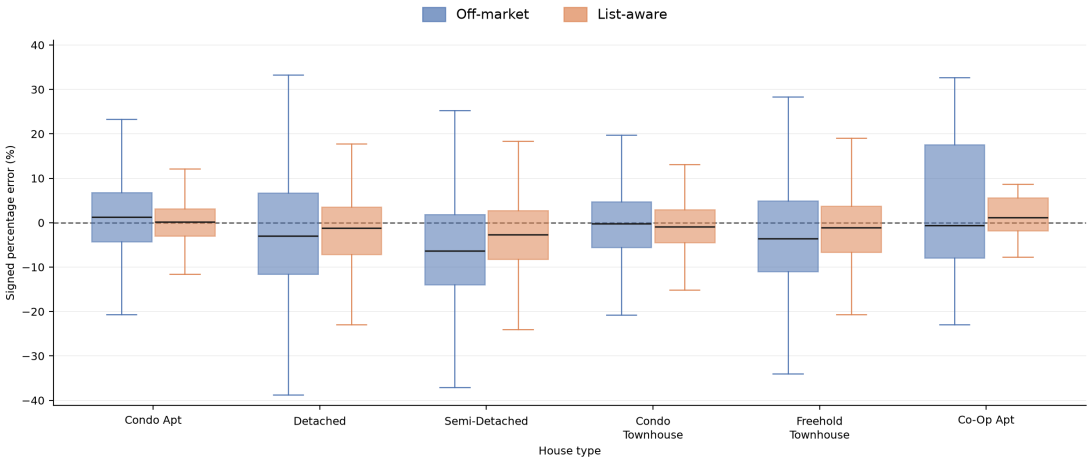


Figure 2: Signed percentage error by house type.

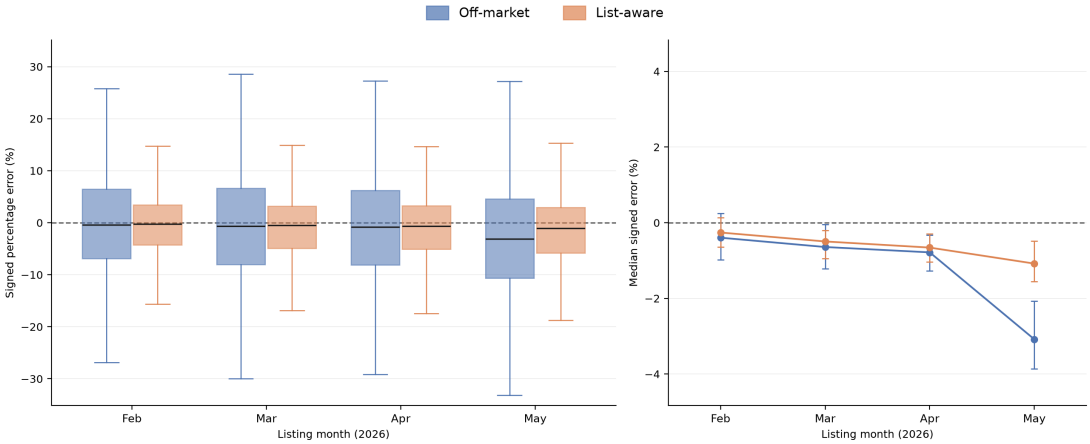


Figure 3: Signed percentage error by month.

partly attributed to the fact there lacks close comparable sales. When the list price is available, our model does a good job predicting close price. Additionally, the model is less accurate as the listing month gets further away from the cutoff month.

Outside data

Source	URL	Periods covered
None		

Reproducing our results

```
pip install -r requirements.txt
python run.py predict
python run.py check
```

AI Usage Disclosure. Generative AI tools were used in this project for coding assistance and brainstorming.